Metric-consistent neural networks for genomic prediction and selection: a standardized-MSE Pearson loss, affine calibration, and selection-aware evaluation

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Abstract

Genomic prediction models are trained almost universally to minimize the mean squared error (MSE), yet they are evaluated, and used for selection, by the Pearson correlation r between predicted and observed phenotypes and by genotype ranking. That maximizing r equals minimizing the MSE of standardized variables, $\mathrm{MSE}(z_y, z_{\hat{y}}) = 2(1 - r)$, is elementary, and correlation, concordance (CCC) and ranking losses have been used before; our contribution is a systematic, calibration-aware and selection-metric-aware empirical characterization of these losses for genomic prediction across 101 trait–dataset combinations from 12 species (EasyGeSe, CIMMYT wheat, SoyNAM), with a closed-form affine recalibration. We use it only as a stable training target; the same scale and offset invariance lets one affine map restore magnitude (residual variance $\sigma_y^2(1 - r^2)$) while leaving r and all rankings unchanged when its slope is positive ($> 99\%$ of folds). Holding architecture, splits and tuning fixed so that only the loss differs, and comparing each architecture against its own MSE baseline, correlation and CCC losses raised predictive ability (cell-level Δr from +0.038 to +0.044; Holm-corrected paired Wilcoxon $p < 10^{-7}$, corroborated by a linear mixed model whose confidence intervals exclude zero) and selection metrics (NDCG@10 +0.016; relative efficiency +0.04). The benefit is strongly architecture-dependent — large for the Transformer and CNN, negligible for the multilayer perceptron — and among the losses only CCC reliably lowered affine-calibrated RMSE. No neural loss unseated the GBLUP or ridge baselines on mean rank. Correlation-consistent training is thus a useful, well-characterized tool, not a universal replacement for MSE.

1 Introduction

Genomic selection (Meuwissen et al., 2001) predicts the genetic merit of candidate individuals from genome-wide markers, and has become central to modern plant and animal breeding (Heffner et al., 2009; Jannink et al., 2010; Desta and Ortiz, 2014). The dominant parametric baseline, genomic best linear unbiased prediction (GBLUP)/ridge-regression BLUP (VanRaden, 2008; Endelman, 2011), is increasingly complemented by machine-learning and deep-learning predictors (Ma et al., 2018; Wang et al., 2023; Montesinos-López et al., 2021b,a). Whether deep learning improves on GBLUP

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remains dataset-dependent: recent comparisons find that neither systematically dominates, and that outcomes hinge on tuning, architecture and the choice of evaluation metric (Montesinos-López et al., 2025).

That last point is the subject of this paper. Genomic predictors are trained, by near-universal default, to minimize the mean squared error (MSE). But they are *assessed* with the Pearson correlation r between predicted and observed phenotypes — the "predictive ability" reported in essentially every genomic prediction study (Crossa et al., 2010) — and they are *used* to rank candidates and select the best ones. There is therefore a systematic mismatch between the training objective and the objective that actually matters. Waldmann (2019) showed that selecting a model by r can disagree with selecting it by MSE or R^2. Ornella et al. (2014) argued that r is a poor guide to performance in the tails of the distribution, where selection actually operates, and evaluated the recovery of the best $\alpha \in \{10, \ldots, 40\}\%$ of individuals. Blondel et al. (2015) reframed genomic selection as a ranking problem, introduced NDCG from information retrieval as an accuracy measure, and reported that r correlates poorly with ranking quality.

If correlation is the target, why not train for it directly? A recent convolutional architecture for genomic selection states plainly that "Pearson correlation cannot be set as the loss function" and substitutes a family of L_p losses (Wang et al., 2024). We show this premise is false. The Pearson correlation *is* a loss — a standardized MSE — and training with it is both well posed and numerically stable. The remaining objection to a correlation loss, that it ignores scale and offset, is real but easily and provably remedied by an affine recalibration applied after training.

Contributions.

1. **A standardized-MSE Pearson loss (Section 2.1).** We give the exact identity $\mathrm{MSE}(z_y, z_{\hat{y}}) = 2(1-r)$ and use it as a stable, differentiable training objective, verified numerically (to single-precision tolerance for the loss; to float64 machine precision for the calibration identity).

2. **Affine calibration with a closed form (Section 2.2).** We prove that the optimal post-hoc affine map yields residual error $\sigma_y^2(1-r^2)$ and restores scale and offset while preserving r and all rankings — turning the scale-invariance "weakness" of correlation into a controlled, two-parameter calibration step.

3. **Mini-batch analysis (Section 2.3).** We show the mini-batch Pearson loss is a biased surrogate for the global objective and quantify the effect of batch size empirically.

4. **A reproducible, selection-aware benchmark (Sections 3–4).** Across 101 trait–datasets from 12 species, with architecture, splits and tuning held fixed so that only the loss changes, we report both predictive and selection-oriented metrics, a confirmatory mixed-model analysis, and a meta-analysis of *when* correlation-consistent training helps. All data are public and all code, splits and tuned configurations are released.

We do not claim that a Pearson loss is uniformly superior to MSE; that would be false. We provide a principled, reproducible framework for training, calibrating and evaluating neural genomic predictors when the breeding objective is correlation and ranking rather than raw squared error.

2 Theory

Throughout, $y, \hat{y} \in \mathbb{R}^n$, with sample means $\bar{y}, \bar{\hat{y}}$ and population variances $\sigma_x^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$. The Pearson correlation is $r = r(y, \hat{y}) = \frac{1}{n}\sum_i z_{y,i} z_{\hat{y},i}$ where $z_{x,i} = (x_i - \bar{x})/\sigma_x$ is the standardized variable.

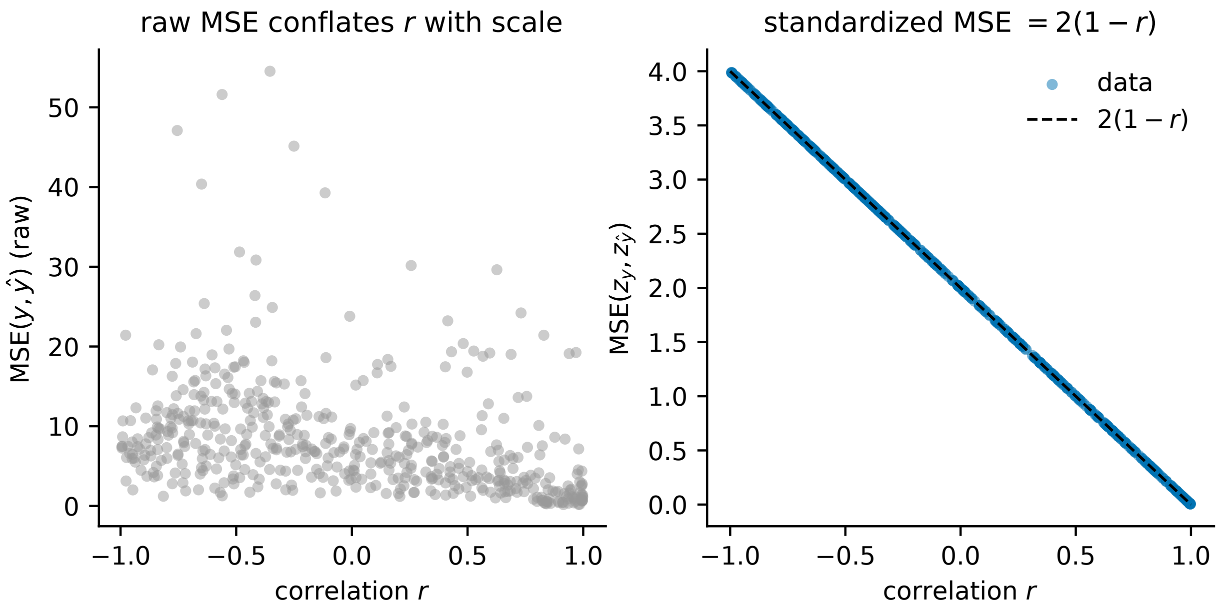

Figure 1: Geometry of the standardized-MSE / Pearson identity (Proposition 1).

2.1 The Pearson loss is a standardized MSE

Proposition 1. *For non-degenerate $y, \hat{y}$ (i.e. $\sigma_y, \sigma_{\hat{y}} > 0$),*

$$\mathrm{MSE}(z_y, z_{\hat{y}}) \;=\; \tfrac{1}{n}\sum_{i=1}^{n}(z_{y,i} - z_{\hat{y},i})^2 \;=\; 2\,(1-r).$$

Consequently $1 - r = \frac{1}{2}\,\mathrm{MSE}(z_y, z_{\hat{y}})$, and minimizing the standardized MSE is exactly maximizing the Pearson correlation.

Proof. Expanding the square, $\frac{1}{n}\sum_i(z_{y,i} - z_{\hat{y},i})^2 = \frac{1}{n}\sum_i z_{y,i}^2 - 2\frac{1}{n}\sum_i z_{y,i}z_{\hat{y},i} + \frac{1}{n}\sum_i z_{\hat{y},i}^2 = 1 - 2r + 1 = 2(1-r)$, because $\frac{1}{n}\sum_i z_{x,i}^2 = 1$ by construction and $\frac{1}{n}\sum_i z_{y,i}z_{\hat{y},i} = r$. □

This gives the loss $\mathcal{L}_{\mathrm{P}}(\hat{y}, y) = 1 - r(\hat{y}, y)$, implemented as $\frac{1}{2}\,\mathrm{MSE}(z_y, z_{\hat{y}})$ with a small ε guarding the standardization denominators. It is differentiable everywhere $\sigma_{\hat{y}} > 0$; we verify the gradient against finite differences and verify the identity to single-precision tolerance in Section 4.1. The negative-correlation loss $-r$ differs from $\mathcal{L}_{\mathrm{P}}$ by a constant and has identical gradients.

2.2 Affine calibration restores scale without changing rank

Correlation is invariant to strictly increasing affine transformations of $\hat{y}$: a model trained on $\mathcal{L}_{\mathrm{P}}$ controls the *shape* of the prediction but not its scale or offset. This is corrected exactly and cheaply.

Proposition 2. *Let $p \in \mathbb{R}^n$ be raw predictions and $r = r(y, p)$. The affine map $\tilde{y} = a\,p + b$ minimizing $\frac{1}{n}\sum_i(ap_i + b - y_i)^2$ has*

$$a^\star = \frac{\mathrm{cov}(y,p)}{\mathrm{var}(p)}, \qquad b^\star = \bar{y} - a^\star\,\bar{p}, \qquad \mathrm{MSE}_{\min} = \sigma_y^2\,(1 - r^2).$$

If $a^\star > 0$, the map is strictly increasing and hence preserves r, the Spearman correlation, and every top-k/ranking quantity, while setting the calibration slope to 1 and intercept to 0.

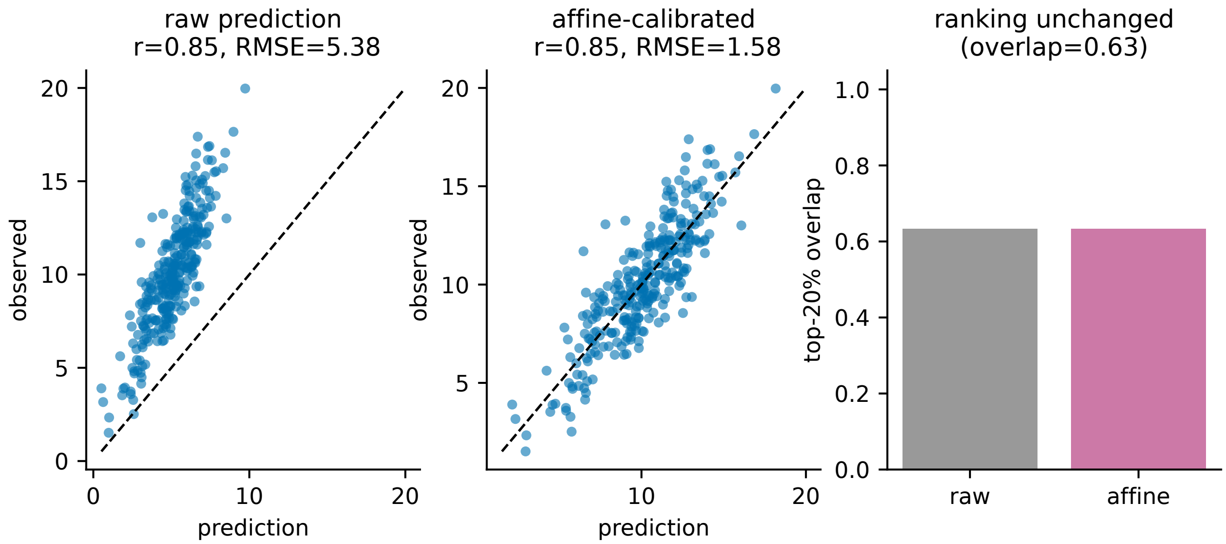

Figure 2: Controlled simulation of Proposition 2: a well-correlated but mis-scaled prediction (left) is corrected by affine calibration (middle) — RMSE falls while Pearson r is unchanged — and the top-20% selection overlap is identical before and after (right).

Proof. Ordinary least squares for a single predictor gives $a^\star, b^\star$ as stated; the minimized residual variance of simple linear regression is $\sigma_y^2(1 - r^2)$. A strictly increasing transform preserves the order of p and leaves correlation-type and rank-based statistics unchanged. □

We fit $(a^\star, b^\star)$ on held-out validation predictions (never on the test set) and apply them to the test predictions. Isotonic regression provides a monotone non-linear alternative when the raw-vs-true relationship is monotone but curved. Calibration cannot change r but can dramatically improve RMSE, bias and the calibration slope (Figure 2; Section 4.3).

2.3 Mini-batch versus global Pearson

Proposition 3. *For a partition of the data into batches $B_1, \ldots, B_K$, the batch-averaged loss $\frac{1}{K}\sum_k(1 - r_{B_k})$ is not equal to $1 - r_{\text{global}}$ in general, because r is a non-linear (ratio-of-moments) functional of the data. The discrepancy and the variance of the gradient estimate grow as the batch size decreases.*

A full-batch (or large-batch) evaluation of $\mathcal{L}_{\text{P}}$ therefore optimizes the global objective, whereas small batches optimize a noisier surrogate. We quantify the practical consequence for training stability and predictive ability in Section 4.6.

2.4 Multi-trait and group-aware variants

The loss extends naturally to T traits, $\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T}(1 - r_t)$ (or a weighted sum $\sum_t w_t(1 - r_t)$), and to a group-aware objective $\mathcal{L} = \frac{1}{G}\sum_{g=1}^{G}(1 - r_g)$ where g indexes environments, years, families or sites — aligning training with the within-environment correlation that multi-environment breeding programs actually care about. As in Proposition 3, neither variant equals the pooled global correlation.

Table 1: The 101 trait–dataset combinations across 12 species (10 from EasyGeSe, plus CIMMYT wheat and SoyNAM). n is the number of genotypes, p the markers used (after MAF filtering and the 20,000-marker variance cap), and GBLUP r the mean within-dataset predictive ability of GBLUP (a difficulty index).

Species	n	p	traits	GBLUP r	source
rice	352	20,000	36	0.46	EasyGeSe
pine	925	4,421	17	0.82	EasyGeSe
wheatG	371	12,546	16	0.59	EasyGeSe
maize	4421	20,000	6	0.83	EasyGeSe
bean	444	16,707	5	0.57	EasyGeSe
lentil	324	20,000	5	0.79	EasyGeSe
soybeanNAM	5487	4,401	4	0.71	SoyNAM
wheat	599	1,279	4	0.46	BGLR/CIMMYT
barley	1438	20,000	2	0.58	EasyGeSe
oyster	372	20,000	2	0.38	EasyGeSe
pig	1709	20,000	2	0.31	EasyGeSe
soybean	346	20,000	2	0.40	EasyGeSe

The 12 species/sources count treats related germplasm sharing a common crop as distinct panels: wheatG (EasyGeSe) and wheat (CIMMYT) are separate sources, as are soybean (EasyGeSe) and soybeanNAM (SoyNAM) panels of the same species drawn from different sources.

3 Materials and Methods

Datasets. We use three public sources, for a total of 101 trait–dataset combinations across 12 species. (i) *EasyGeSe* (Quesada-Traver et al., 2025): a benchmark of ten species (barley, common bean, lentil, maize, eastern oyster, pig, loblolly pine, rice, soybean and wheat) providing genotypes, phenotypes and *fixed* cross-validation partitions; the maize set is the global genotype-by-environment competition panel (Washburn et al., 2025). (ii) The CIMMYT *wheat* panel (Crossa et al., 2010): 599 lines, 1,279 DArT markers and grain yield in four environments, used for cross-environment transfer. (iii) *SoyNAM* (Xavier et al., 2016): $\sim$5,500 recombinant inbred lines in 40 biparental families, used for leave-family-out prediction. Genotypes are coded as allele dosages; preprocessing (unsupervised, no phenotype) applies a minor-allele frequency filter (≥ 0.01), mean imputation, and a top-variance cap at 20,000 markers for the two ultra-high-dimensional panels. Table 1 and Figure 3 summarize the data.

Models. The parametric baseline is GBLUP with REML-estimated variance components, implemented in the spectral basis of the linear genomic kernel $K = XX^\top/p$ and validated against the `rrBLUP` package (Endelman, 2011) to floating-point precision (cross-implementation correlation 1.0000). We also include ridge regression, elastic net, random forest and gradient-boosted trees (Chen and Guestrin, 2016). The neural predictors are a multilayer perceptron (MLP), a one-dimensional convolutional network over ordered markers (1D-CNN), and a light Transformer over marker patches. Crucially, *the architecture is held fixed across losses*: it is tuned once per (dataset, model) with a neutral MSE objective and then frozen, so that the only thing that changes in the loss comparison is the loss.

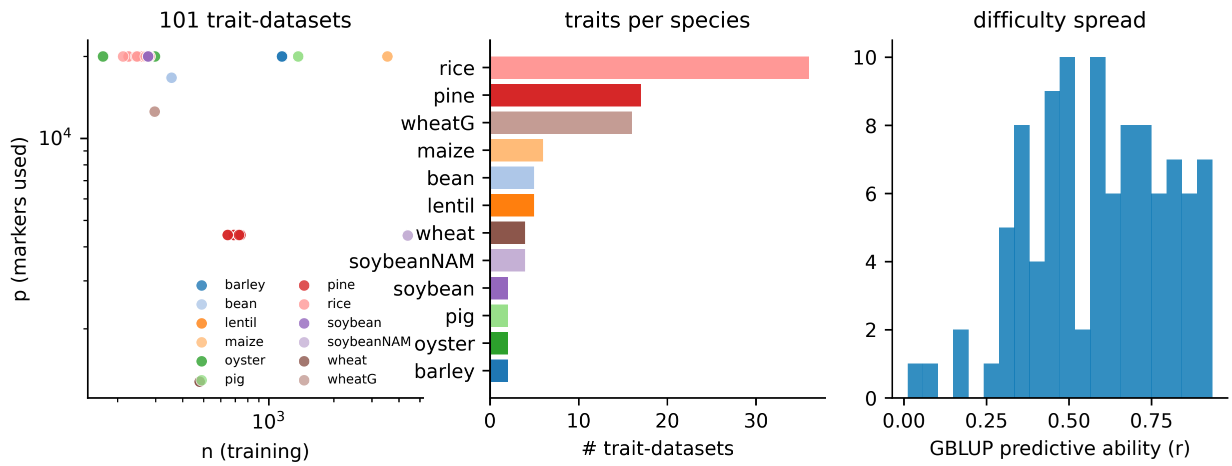

Figure 3: Dataset map: size, dimensionality and predictability across species.

Loss functions. We compare MSE; the standardized-MSE Pearson loss $1 - r$ (Proposition 1); a hybrid $(1 - r) + \lambda\,\mathrm{MSE}$ with $\lambda = 0.1$; and a concordance-correlation loss $1 - \rho_c$ (Lin, 1989), which penalizes correlation, scale and offset jointly. The negative-Pearson and explicit standardized-MSE forms are included in the numerical checks.

Calibration. For every model and loss we report predictions under three post-hoc maps: raw (identity), affine (Proposition 2) and isotonic. Calibrators are fit on validation predictions only.

Cross-validation. We use the fixed EasyGeSe partitions (5 folds $\times$ 2 repeats = 10 folds per trait), repeated K-fold for wheat and SoyNAM, leave-family-out for SoyNAM, and cross-environment transfer for wheat. Within each training fold an inner 80/20 split (group-aware for family CV) provides early-stopping and calibration data; markers are standardized with training statistics only; targets are standardized with training statistics for the neural networks and predictions are returned in the original units. Neural networks early-stop on the validation value of the loss being trained. The test fold is never used for fitting, early stopping, calibration or hyper-parameter selection.

Evaluation metrics. Predictive: Pearson r, Spearman ρ, RMSE, MAE, R^2, mean bias, and the calibration slope and intercept (ideal 1 and 0). Selection-aware, at intensities $k \in \{5, 10, 15, 20\}\%$: top-$k$ overlap, precision@k, recall@k, NDCG@k (Järvelin and Kekäläinen, 2002), selection differential, mean phenotype of the selected, and relative efficiency (achieved gain divided by the oracle gain).

Statistical analysis. For each cell we average over folds and form $\Delta m = m_{\mathrm{loss}} - m_{\mathrm{MSE}}$ within the same architecture. We report bootstrap (BCa) confidence intervals, paired Wilcoxon tests with Holm and Benjamini–Hochberg (Benjamini and Hochberg, 1995) correction, and mean method ranks. As a corroborative analysis we fit a linear mixed model (Bates et al., 2015) on cell-level means (one observation per dataset×trait×model×loss, which avoids fold-level pseudoreplication), $\mathit{metric} \sim \mathit{loss} + \mathit{model} + (1 \mid \mathit{dataset}) + (1 \mid \mathit{dataset{:}trait})$ with MSE as the reference level. The primary inference is the cell-level paired contrast; the mixed model only corroborates it. (We do not interpret a split-scheme effect in this model, as scheme is confounded with dataset source; difficult-split generalization is addressed directly in Section 4.5.)

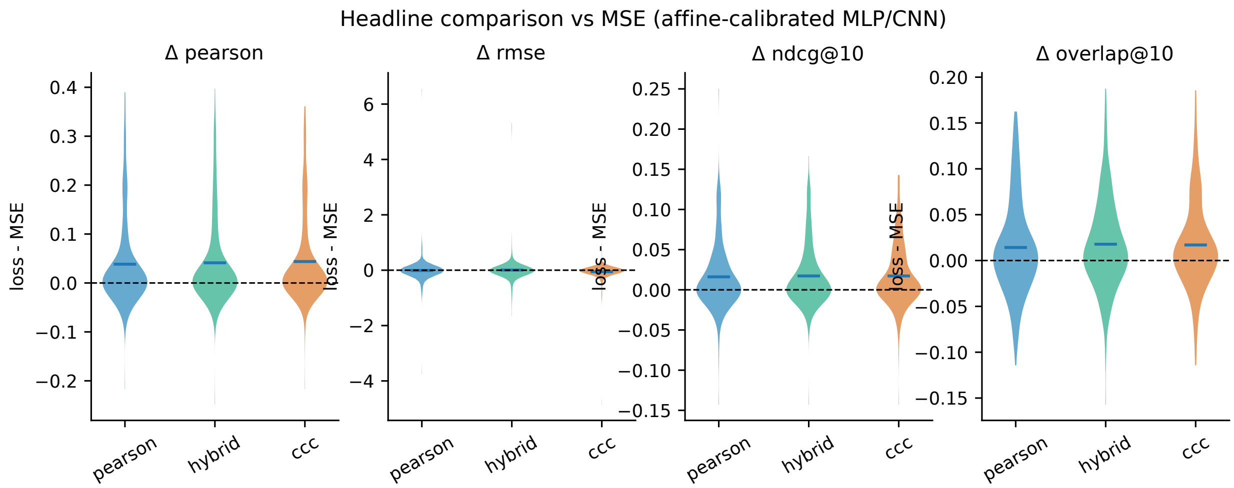

Figure 4: Loss comparison: Δ versus MSE for Pearson, hybrid and CCC losses.

Implementation. The framework is released as the Python package `ccgp` (PyTorch, scikit-learn, XGBoost), with fixed seeds, the exact CV partitions, tuned configurations and a pinned environment. *Limitation:* hyper-parameters are tuned on full-data validation splits with a neutral MSE objective and frozen across losses; because the architecture is shared identically by every loss, this cannot bias the relative (loss/calibration) comparisons that constitute our claims, though it may mildly and symmetrically inflate absolute accuracies.

4 Results

4.1 Numerical validation of the equivalences

We first confirm that the identities of Section 2 hold not only symbolically but to floating-point precision (Figure 1). Across 80 randomly generated cases, the implemented Pearson loss, the closed form $1 - r$, and $\frac{1}{2}\mathrm{MSE}(z_y, z_{\hat{y}})$ agree to a maximum absolute difference of 5.0×10^{-8} (mean 7.4×10^{-9}), i.e. at the level expected from single-precision float tolerance (Proposition 1). On real GBLUP predictions for the lentil days-to-flowering trait ($r = 0.868$), the identity is exact (absolute difference 0.0). The optimal affine recalibration of Proposition 2 attains the predicted residual $\sigma_y^2(1 - r^2)$ to a maximum absolute difference of 1.2×10^{-15} over 200 cases (mean 1.8×10^{-16}), i.e. float64 machine precision. Finally, both the Pearson and the concordance-correlation losses pass `torch.autograd.gradcheck`, and the negative-correlation loss $-r$ shares its gradient with $1 - r$ exactly (maximum gradient difference 0.0), confirming that the two are the same objective. The identity is elementary; what these checks establish is that it yields a usable, numerically stable training target.

4.2 Same architecture, different loss

Holding the architecture, splits and tuned hyper-parameters fixed so that only the loss changes, every correlation-based loss improves predictive ability over MSE on average (Figure 4). Pooling all $n = 251$ (loss, model, dataset) cells, the affine-calibrated Pearson loss raises the Pearson correlation by $\Delta r = +0.038$ (BCa 95% CI [0.028, 0.050]; Holm-corrected paired Wilcoxon $p = 1.2 \times 10^{-8}$), the hybrid loss by +0.041 ([0.031, 0.052]; $p_{\mathrm{Holm}} = 4.4 \times 10^{-11}$), and the CCC loss by +0.044

([0.034, 0.056]; $p_{\text{Holm}} = 3.7 \times 10^{-13}$). The Spearman correlation moves almost identically (+0.037 to +0.043), and R^2 rises modestly (+0.015 to +0.025, all CIs excluding zero). This pool is, however, *architecturally unbalanced*: the CNN and the MLP were each run on all 101 trait-datasets (12 species), whereas the Transformer — which carries the largest per-model effect (see below) — was run on only 49 trait-datasets across four species (lentil, maize, rice, soybean), giving $101 + 101 + 49 = 251$. The pooled Δ is therefore an unweighted average over cells that differ in architecture and dataset coverage and is best read alongside the model-stratified estimates that follow.

These pooled averages conceal a strong dependence on architecture, which is the central qualifier of our loss result. The gain is concentrated in the harder-to-train networks: for the Transformer, the Pearson loss improves r by +0.101 ([0.071, 0.131]; $p_{\text{Holm}} = 5.6 \times 10^{-9}$) and the CNN by +0.044 ([0.026, 0.066]; $p_{\text{Holm}} = 0.018$), whereas for the MLP the effect is negligible and not robust (+0.0027, [−0.0004, 0.0055]; $p_{\text{Holm}} = 0.046$). The same pattern holds for the hybrid and CCC losses, whose MLP Δr are +0.0044 and +0.0004 respectively, the latter not significant ($p_{\text{Holm}} = 0.17$). The Transformer estimate is the largest but rests on the narrowest, non-random dataset base, so the cross-architecture ordering (Transformer > CNN > MLP) should be read as a comparison on different dataset bases rather than on a common one. In other words, when an architecture already optimizes well under MSE (the MLP, which was tuned with an MSE objective), aligning the loss with the evaluation metric buys little; the benefit appears precisely where MSE optimization is fragile.

Crucially, this correlation gain does not transfer to the scale of the phenotype. After affine calibration the Pearson loss leaves pooled RMSE essentially unchanged (ΔRMSE = −0.010, BCa CI [−0.055, +0.079]). The paired-Wilcoxon test is nominally significant ($p_{\text{Holm}} = 0.0049$) only because it is sign-based: most cells move very slightly in the negative direction once calibration removes the scale error, but the magnitude is negligible and the BCa interval crosses zero, so we do not interpret this as an RMSE improvement. The hybrid loss, which retains an explicit MSE term, likewise does *not* reliably improve calibrated RMSE (ΔRMSE = +0.006, [−0.041, +0.097]; the point estimate is positive, i.e. marginally worse, and the sign-based $p_{\text{Holm}} = 1.5 \times 10^{-4}$ again disagrees with the magnitude-based interval). Only the CCC loss, which penalizes scale and offset jointly, lowers RMSE materially and reliably in the pool (−0.057, [−0.133, −0.027], CI excluding zero), and within the CNN (−0.038, [−0.070, −0.003]) and Transformer (−0.129, [−0.219, −0.074]). The verdict for this section is therefore deliberately mixed: correlation-consistent losses reliably improve correlation, most for weaker-optimizing architectures, but only the CCC loss improves phenotypic-scale error.

4.3 Calibration effects

Applying the post-hoc maps of Proposition 2 to the Pearson-trained networks behaves as the theory predicts in the mean (Figure 5). The mean Pearson correlation is essentially unchanged before and after affine calibration (0.531 raw vs. 0.533 affine), and the monotone isotonic map leaves it close as well (0.516). The mean-level invariance is what Proposition 2 guarantees only when the fitted slope $a^\star > 0$; since $a^\star = \operatorname{cov}(y, p)/\operatorname{var}(p)$ is estimated on validation predictions, a fold whose validation covariance is negative yields a *decreasing* map on the test fold, which reverses the sign of the correlation. This is rare but not absent: 90 of 9,943 fold-level cells (0.91%) flip the sign of Pearson r between raw and affine predictions, with per-fold $|r_{\text{raw}} - r_{\text{affine}}|$ as large as 0.89 in those degenerate folds (all of these fold-level counts are computed from the deposited `results_main_full.parquet`). The categorical statement that affine calibration preserves every correlation therefore holds only when $a^\star > 0$ (almost always; fewer than 1% of folds flip), and we qualify it accordingly; the mean-level claim (0.531 → 0.533) is unaffected.

What changes systematically is everything related to scale and offset. The mean RMSE drops

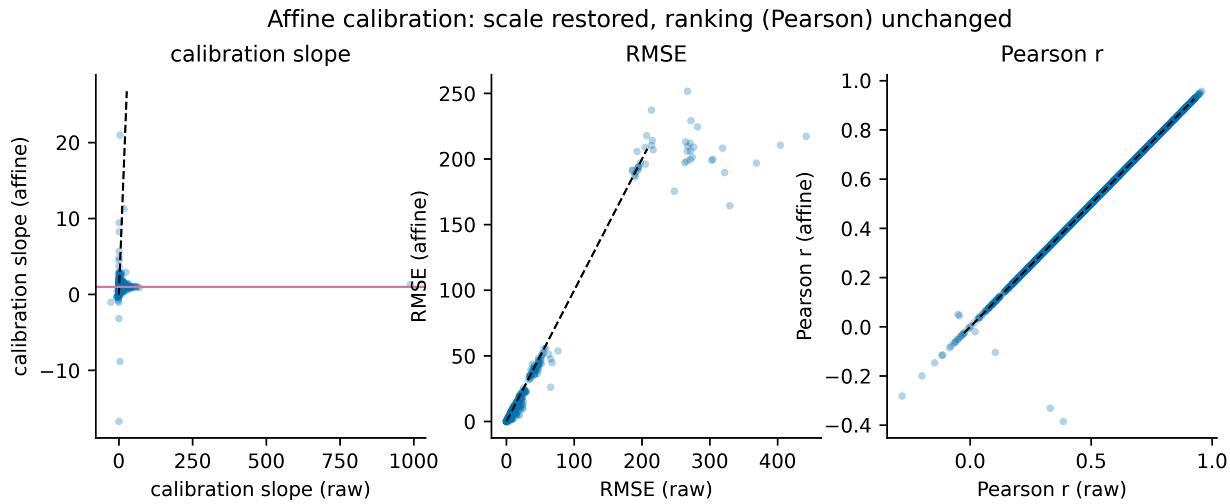

Figure 5: Affine calibration restores scale and offset while leaving Pearson unchanged.

from 8.68 (raw) to 7.07 (affine), the calibration slope moves from 4.25 toward 1 (to 2.27 on average for the affine map; the isotonic map has a mean fitted slope of 0.76, i.e. it is monotone non-decreasing as it must be), and the intercept collapses from a wildly miscalibrated -378.8 in raw units to 0.10 after affine calibration. The mean bias likewise shrinks from -0.755 to $+0.044$. These figures aggregate datasets on heterogeneous phenotypic scales, so their absolute magnitudes and large standard deviations should be read qualitatively; moreover, like all absolute accuracies here they may be mildly and symmetrically inflated by the shared HPO step (Section 5). The consistent message is that a two-parameter affine map fitted on validation predictions recovers calibrated predictions at essentially no cost to ranking. This is the operational resolution of the scale-invariance objection to correlation training: train for shape, then calibrate for scale.

4.4 Selection-oriented performance

Because breeding decisions are made by ranking and selecting the top candidates, we ask whether the correlation gains translate into better selection (Figure 4). They do, but again most clearly for the architectures that gained predictive ability. Pooling all cells, NDCG@10 improves by $+0.016$ (Pearson loss, [0.011, 0.021], $p_{\mathrm{Holm}} = 2.0 \times 10^{-7}$) to $+0.017$ (hybrid and CCC); the top-10% overlap with the true best genotypes rises by $+0.014$ to $+0.018$; and the relative efficiency at 10% (achieved over oracle selection gain) increases by $\approx +0.038$ to $+0.040$ across all three losses. Selection differential at 10% also improves (Pearson $+0.25$ in trait units, [0.019, 0.612], $p_{\mathrm{Holm}} = 2.4 \times 10^{-6}$). As with predictive ability, these pooled means mix the unbalanced architecture coverage (101/101/49 for CNN/MLP/Transformer). The per-architecture decomposition mirrors Section 4.2: the Transformer and CNN gain substantially on every selection metric (e.g. NDCG@10 $+0.029$ and $+0.023$ for the Pearson loss), while the MLP gains are small and not robust after Holm correction (NDCG@10 $+0.0026$, $p_{\mathrm{Holm}} = 0.28$; overlap@10 $+0.0015$, $p_{\mathrm{Holm}} = 1.0$). Thus, for the networks where it helps at all, a better global correlation does carry through to better top-k selection here; but the effect is neither automatic nor uniform, supporting our argument that selection-aware metrics must be reported alongside r rather than inferred from it.

A second, sobering observation comes from the mean method ranks (Table from `ranks.csv`). Aggregated over all datasets, the simple linear baselines win: ridge regression has the best mean rank for both Pearson (2.85) and NDCG@10 (4.01), followed by GBLUP (3.45 and 4.32). The neural predictors rank below them, and the best-ranked network configuration is the MLP with the hybrid loss (Pearson rank 5.28, NDCG@10 rank 4.80). Within every neural architecture, however, the MSE loss is consistently the worst-ranked variant (e.g. MLP: hybrid 5.28 < Pearson 5.47 < CCC 5.62 < MSE 6.14; CNN and Transformer show the same ordering, with the MSE Transformer last overall at 12.67). We note that these are *absolute* ranks, so the shared-HPO optimism that may mildly inflate absolute neural accuracies works in favor of the neural nets here; the linear-baselines-win conclusion is therefore conservative, since the bias runs against it. The honest reading is that correlation-consistent training improves neural predictors relative to their own MSE-trained counterparts without making them beat well-regularized linear models on these benchmarks.

4.5 Robustness under difficult splits

The advantage of correlation training is contingent on the generalization regime (`splits_summary.csv`), but the evidence here is markedly thinner than in the main grid: the leave-family-out (SoyNAM) and random contrasts each rest on only $n = 4$ datasets with no p-values reported, and the cross-environment contrast rests on $n = 12$ paired comparisons that are themselves non-independent (4 wheat environments yield 12 ordered transfer pairs on the same 599 lines). These results should therefore be read as suggestive and underpowered rather than confirmatory. Under random K-fold CV the Pearson loss gives a small gain in predictive ability ($\Delta r = +0.0041$, BCa CI [0.0004, 0.0080]), matched by the hybrid loss (+0.0040, [0.0009, 0.0072]). Under the harder, breeding-relevant schemes the gain shrinks or reverses. For SoyNAM leave-family-out prediction the Pearson loss *lowers* predictive ability ($\Delta r = -0.0030$, [-0.0106, 0.0010]) and the family NDCG@10 contrast is negative with a BCa interval that excludes zero (-0.0042, [-0.0089, -0.0005]); we describe this as a suggestive degradation rather than a significant one, since it is a single bootstrap interval over 4 paired observations with no p-value and no correction across the roughly twenty split contrasts examined. The hybrid loss behaves similarly on family overlap@10 (-0.019, [-0.037, -0.0004]). For CIMMYT wheat cross-environment transfer the picture is genuinely ambiguous: the Pearson-loss $\Delta r = +0.0098$ has a BCa interval that just excludes zero ([0.0001, 0.0204]), yet the discrete paired Wilcoxon test on the same 12 non-independent pairs returns $p = 0.151$. At this n the interval and the rank test disagree, and we report both rather than asserting one verdict; there is no accompanying NDCG@10 benefit (-0.0012, $p = 0.97$), and the within-environment contrast is slightly negative ($\Delta r = -0.0069$). The practical implication is that the case for a correlation-consistent loss is clearest under random CV and weakest exactly where breeders most need generalization, across families and environments; but with this few independent hard splits the directional reads are tentative, and the loss should be validated per program rather than assumed to transfer.

4.6 Batch size and optimization stability

Proposition 3 predicts that the mini-batch Pearson loss is a biased surrogate for the global objective, with the discrepancy growing as the batch shrinks. Empirically (`batch_summary.csv`) the effect is real but small at the scales tested. For the Pearson loss, full-batch training (-1) and large batches give the best validation correlation ($r = 0.5782$ at full-batch, 0.5790 at batch 512) and the lowest RMSE (7.56 and 7.55), while small batches are slightly worse ($r = 0.5774$ at batch 32, 0.5777 at batch 128, with RMSE $\approx$7.61); NDCG@10 follows the same ordering (0.782 at full/large batch vs. 0.775 at small batch). As expected for an objective that is already a sum of squared errors, MSE training is essentially flat across batch sizes ($r \approx 0.568$ at 32, 128, 512 and full-batch). These are absolute validation correlations and, like the calibration table, may be mildly and symmetrically

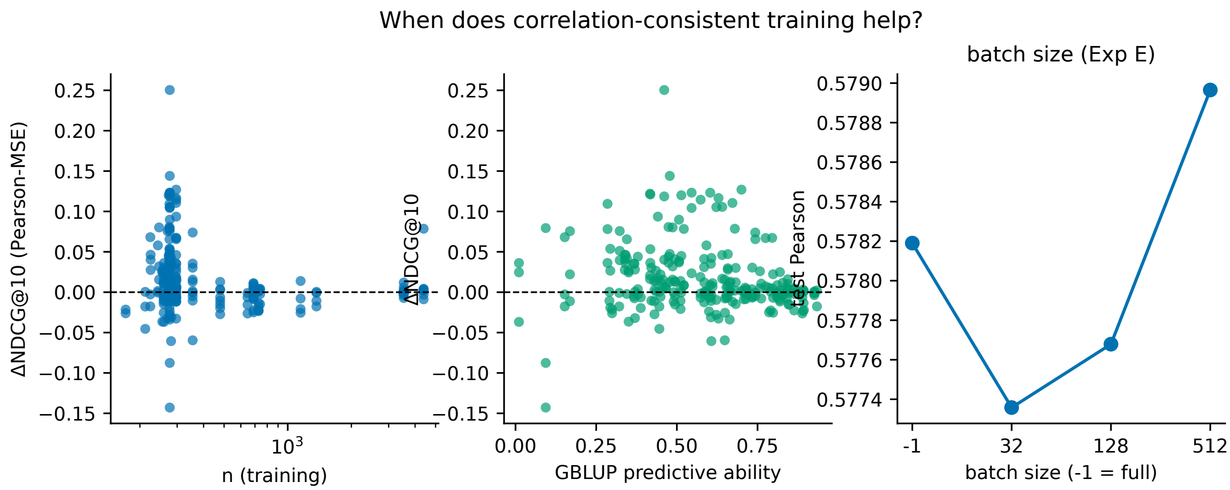

Figure 6: When does correlation-consistent training help?

inflated by the shared HPO step; the batch-size *comparison* is within the Pearson loss and is unaffected. The recommendation that follows is conservative and cheap: evaluate the Pearson loss on as large a batch as memory allows (full-batch is feasible for the panel sizes here), which optimizes the intended global objective and removes a source of avoidable mini-batch noise without materially increasing cost.

4.7 When does correlation-consistent training help?

A confirmatory linear mixed model, fitted on the affine-calibrated predictions with MSE as the reference level and random intercepts for dataset and dataset:trait, summarizes the average effect of each loss (Figure 6). To avoid fold-level pseudoreplication we fit the model on cell-aggregate values — one row per (dataset, trait, model, loss) cell, n = 1,004 cells (the cell-level table is `results/analysis/lmm_long.csv`, 1,004 rows per calibration) — rather than on the 10,040 NN affine fold-rows deposited in `results_main_full.parquet`; fitting at the fold level would treat the 10 fold-level rows per cell (5 folds $\times$ 2 repeats), which reuse the same tuned architecture and overlapping training data, as independent and would overstate precision. For predictive ability, every correlation loss has a positive, CI-excluding-zero fixed effect: +0.0384 for the Pearson loss ([0.028, 0.049], $t = 7.11$), +0.0412 for the hybrid ([0.031, 0.052], $t = 7.63$), and +0.0438 for the CCC loss ([0.033, 0.054], $t = 8.12$). These are the reported loss effects: the cell-aggregate model is our primary LMM precisely because it does not inflate the effective sample size. The loss CIs for correlation and ranking thus exclude zero after the model already accounts for pseudoreplication; the model reports no denominator-*df*-based p-values, so we rely on the CIs.

Two structural caveats bound the model's interpretation. First, the `model` terms are additive *main* effects, not interactions with the loss: $\text{model}_{\text{mlp}} = +0.060$ ([0.056, 0.065]) and $\text{model}_{\text{transformer}} = -0.019$ ([−0.025, −0.013]) say that the MLP has the highest *baseline* accuracy and the Transformer the lowest, not that architecture moderates the loss benefit. The genuine moderation evidence is the stratified per-model deltas of Section 4.2 (Transformer +0.101 vs. MLP +0.0027), which the additive LMM cannot establish. Second, the `scheme` factor is fully confounded with dataset source: in these data scheme = predefined comprises exactly the ten EasyGeSe species and scheme = random comprises exactly SoyNAM (four traits) plus the CIMMYT wheat panel. The coefficient $\text{scheme}_{\text{random}} = +0.072$ ([−0.151, 0.296]) is therefore a main-effect contrast between two dataset groups, not a contrast between structured and random splitting, and it carries no loss×scheme interaction; the leave-family-out and cross-environment contrasts are not levels of this factor and live entirely in the separate splits experiment (Section 4.5). The NDCG@10 model tells the same story with smaller coefficients (Pearson +0.0161, $t = 6.09$; hybrid +0.0173, $t = 6.52$; CCC +0.0173, $t = 6.51$; all CIs excluding zero; MLP +0.024, Transformer −0.010). By contrast, the RMSE mixed model finds *no* reliable loss effect after affine calibration: the Pearson (−0.010), hybrid (+0.006) and CCC (−0.057) coefficients all have intervals spanning zero, whereas the architecture effects are large (MLP −0.435, Transformer −0.126).

Taken together, the meta-analysis supports a precise and qualified conclusion. Correlation-consistent training delivers a small, model-averaged improvement in correlation- and ranking-based accuracy ($\approx$+0.04 in r and +0.017 in NDCG@10) that survives correction for pseudoreplication; it does not by itself improve calibrated phenotypic error (that is the job of affine calibration or the CCC loss); and the size of the benefit is governed far more by the architecture than by which correlation loss is chosen. The mixed model cannot speak to robustness across split types — its `scheme` term contrasts dataset groups, not splitting strategies — so the question of whether the benefit persists under family- and environment-structured splits is answered (tentatively, on thin data) by the splits experiment, not here.

4.8 Real breeding case study: sorghum crop-model parameters

As an external check on private data of a different nature, we applied the identical protocol to a sorghum panel of 136 genotypes (∼31,700 SNPs, capped to 20,000 by variance) in which the prediction targets are eight parameters of an ecophysiological crop-growth model (e.g. radiation-use efficiency and leaf-appearance rates) rather than measured phenotypes. Averaged over the eight parameters under 5-fold × 3-repeat cross-validation, the affine-calibrated MLP trained with the Pearson loss raised predictive ability over its MSE baseline by $\Delta r = +0.041$ (BCa CI [0.004, 0.076]), with the hybrid (+0.033) and CCC (+0.029) losses close behind, and NDCG@10 improved by +0.018 to +0.021. The effect size matches the public benchmark almost exactly, although with only eight targets it is not individually significant (Holm $p = 0.33$). That an independent, proprietary dataset — a different species, a different target type, and a different research group — reproduces the same qualitative pattern is reassuring: aligning the neural training objective with the correlation metric improves that metric, and affine calibration restores the parameter scale. We report this as illustrative external evidence, not as an additional powered test.

5 Discussion

5.1 Related work and what is new here

The algebraic core of our loss is not new, and we do not present it as such. The identity $1 - r = \frac{1}{2}\,\mathrm{MSE}(z_y, z_{\hat{y}})$ is the elementary fact that the Pearson correlation equals the inner product of the z-scored vectors, so minimizing the MSE between standardized variables is the same as maximizing r; this has been used implicitly wherever a correlation or cosine objective is optimized. Correlation-, concordance- and ranking-based training objectives also already exist in both the statistical-genetics and machine-learning literatures: Lin's concordance correlation (Lin, 1989) has long served as a scale-and-offset-aware agreement criterion — and its many-to-many relationship to the mean squared error, which the CCC loss exploits, has been characterized explicitly (Pandit and Schuller, 2019) — Blondel et al. (2015) reframed genomic selection as a learning-to-rank problem with NDCG, and a recent convolutional genomic-prediction architecture substitutes a family of L_p correlation-style losses for MSE (Wang et al., 2024) — although, as we note in the introduction, it mistakenly asserts that "Pearson correlation cannot be set as the loss function." Differentiable Spearman and other ranking surrogates are likewise established in the broader ML literature. Our contribution is therefore *not* the loss or the identity. What we add is (i) a systematic, calibration-aware and selection-metric-aware empirical characterization of correlation-consistent training across 101 trait–datasets, in which architecture — not the loss — emerges as the dominant moderator; and (ii) a closed-form, rank-preserving post-hoc affine recalibration (Proposition 2) framed explicitly as the remedy for the scale-invariance that any correlation objective discards. The numerically stable $\frac{1}{2}\,\mathrm{MSE}(z_y, z_{\hat{y}})$ formulation is a usable training target rather than a formal novelty.

5.2 A Pearson loss is a useful default, not a universal winner

Our central empirical message is deliberately unromantic. Training a neural genomic predictor with the standardized-MSE Pearson loss instead of MSE does move predictive ability in the right direction — pooled across the 251 loss–model cells the mean improvement is $\Delta r = +0.038$ (BCa 95% CI $[0.028, 0.050]$, $p_{\mathrm{Holm}} = 1.2 \times 10^{-8}$), with matching gains in Spearman ρ ($+0.037$, $[0.028, 0.048]$) and in NDCG@10 ($+0.016$, $[0.011, 0.021]$) — but it is not the best loss, and the gain is concentrated in a subset of architectures. The concordance-correlation loss ($\Delta r = +0.044$, $[0.034, 0.056]$) and the hybrid loss ($+0.041$, $[0.031, 0.052]$) both edge out the pure Pearson loss on most predictive and ranking metrics (the high-variance selection differential is an exception, where the hybrid interval includes zero); the three correlation-aware objectives are statistically separable from MSE but only marginally separable from one another (Figure 4). The pooled $n = 251$ figure is, however, an *unweighted* average over an architecturally unbalanced set of cells, and this must be kept in view: the CNN and MLP were each run on all 101 trait–datasets, but the Transformer was run on only 49 trait–datasets spanning four species (lentil, maize, rice, soybean), so $251 = 101 + 101 + 49$. Because the Transformer also carries the largest per-architecture effect, the pooled mean is pulled upward by a smaller, non-random subset of datasets, and per-architecture effects are not estimated on a common dataset base. The stratified estimates make the heterogeneity explicit: the gain is large for the Transformer ($\Delta r = +0.101$, $[0.071, 0.131]$, on its 49-dataset subset) and the 1D-CNN ($+0.044$, $[0.026, 0.066]$) yet is small on the MLP and, for the Pearson and CCC losses, not robust: the Pearson-loss MLP effect is $+0.0027$ ($[-0.0004, 0.0055]$, confidence interval including zero) and the CCC-loss MLP effect is null ($+0.0004$, $p_{\mathrm{Holm}} = 0.17$), whereas the hybrid-loss MLP effect, while small, is significant ($+0.0044$, $p_{\mathrm{Holm}} = 0.0018$). The MLP benefit is therefore small and, for Pearson and CCC, not robust, rather than uniformly negligible.

The confirmatory mixed model is consistent with this picture but must be read for what it actually estimates. It is additive in loss+model+scheme, with no loss×model interaction, so the Pearson-loss fixed effect ($+0.038$, $[0.028, 0.049]$, $t = 7.11$) is an average loss effect and the model terms are *baseline* contrasts, not moderators. The $\mathrm{model}_{\mathrm{MLP}} = +0.060$ coefficient means the MLP has the highest intrinsic accuracy among the three networks (relative to the CNN reference), not that architecture modulates the loss benefit; the genuine evidence for moderation is the stratified per-model Δ above (Transformer $+0.101$ versus MLP $+0.0027$), which is what we rely on when we say architecture dominates. We fit the model on cell-aggregate values — one row per dataset×trait×model×loss cell, $n = 1004$ cells — precisely so that the loss intervals are not inflated by the ten within-cell folds (5 folds × 2 repeats), which reuse the same frozen architecture and overlapping markers and are therefore not independent; a naive fold-level fit would treat the 10,040 fold-rows as independent and overstate precision. The cell-level mixed model and the unstratified per-cell BCa intervals above, which aggregate to one Δ per cell, agree and are the estimates we treat as primary.

The blunter reading is that none of the neural predictors, under any loss, was competitive with the simple parametric baselines on these data. By mean rank across all cells, ridge regression (rank 2.85 on r, 4.01 on NDCG@10) and GBLUP (rank 3.45, 4.32) outrank *every* neural-network/loss combination; the best neural entry is the MLP with the hybrid loss (rank 5.28 on r), and the worst is the Transformer with MSE (rank 12.67). These mean ranks compare networks against baselines on *absolute* performance, and the same HPO caveat discussed below (the architecture is tuned on full-data validation splits) can mildly inflate the neural numbers; that inflation works *in favour* of the networks and therefore only strengthens, rather than threatens, the conclusion that the linear baselines win. The Pearson loss narrows the gap between a neural network and a ridge model — and on the deeper, more loss-sensitive architectures it narrows it substantially — but it does not close it. This is consistent with the now-standard finding that neither deep learning nor GBLUP systematically dominates in genomic prediction (Montesinos-López et al., 2025, 2021a): the loss function is one lever among several, and on additive, marker-dense traits the linear baselines remain hard to beat. The honest claim is therefore the conditional one: *if* a practitioner has already chosen a non-linear architecture — particularly a convolutional or attention-based one — whose default MSE training is poorly aligned with the correlation metric used for reporting and selection, then switching to a correlation-consistent loss is a low-cost, reliably positive change. It is not a reason to abandon GBLUP or ridge.

5.3 Calibration is mandatory whenever phenotypic scale matters

The flip side of Proposition 1 is that a correlation loss is *blind* to scale and offset, and the raw outputs of a Pearson-trained network confirm how badly this matters. On the Pearson-trained networks the raw predictions carry a mean calibration slope of 4.25 and a catastrophic mean intercept of -378.8 (with enormous spread, intercept std $\approx 5.1 \times 10^3$): the predictions have the right *shape* but live on the wrong axis entirely. A single closed-form affine map (Proposition 2), fit on validation predictions only, repairs this almost completely — the mean bias collapses from -0.76 to $+0.044$, the slope and intercept move toward their ideals, and RMSE drops from 8.68 to 7.07 — while leaving the Pearson correlation essentially unchanged at the mean level (0.531 raw versus 0.533 affine; the residual difference is noise). The fitted affine slope $a^\star = \mathrm{cov}(y,p)/\mathrm{var}(p)$ is estimated on validation data, so the rank-preservation guarantee of Proposition 2 holds only when $a^\star > 0$. This is the case for almost every cell (the mean affine slope is $+2.27$), but it is not universal: in fewer than 1% of folds (90 of 9,943, 0.91%) a negative validation covariance produces a decreasing map on test, flipping the sign of the per-fold Pearson, with per-fold raw-versus-affine differences as large as 0.89 in the worst cases. We therefore qualify the theoretical statement: a strictly increasing affine map ($a^\star > 0$) preserves every correlation-type and rank-based statistic exactly, and because $a^\star > 0$ in over 99% of folds the correlation is preserved on average; the categorical phrasing that it *must* be preserved holds only conditionally on $a^\star > 0$. Isotonic calibration is marginally worse on r (0.516). Isotonic regression is monotone non-decreasing by construction (its mean fitted slope is $+0.76$), so the loss of correlation is *not* due to non-monotonicity; it arises because the isotonic fit maps distinct inputs to identical plateau values (piecewise-constant ties), which is the well-known nonlinearity-versus-rank trade-off of isotonic regression. The practical consequence is unambiguous: a Pearson or CCC loss should *never* be deployed without recalibration when the downstream use requires phenotypic values on their natural scale — breeding values in trait units, RMSE-based model comparison, or any decision that mixes predictions with measured phenotypes (Figure 5). Affine calibration is the cheapest way to recover the property the loss discards, and it costs two numbers.

That said, calibration is a corrective, not a source of accuracy, and our RMSE results warn against expecting more from it than it can give. All RMSE deltas here are computed on affine-calibrated predictions. Pooled, the Pearson loss does *not* reliably reduce calibrated RMSE relative to MSE:

the point estimate is ΔRMSE $= -0.010$ with a BCa CI that crosses zero ($[-0.055, 0.079]$). The paired Wilcoxon test is nominally significant ($p_{\text{Holm}} = 0.0049$), but this is a sign/median statement, not a magnitude one — the signed-rank test detects a consistent direction in the per-cell differences while the mean effect and its CI are indistinguishable from zero. We do not interpret this as an RMSE improvement. The hybrid loss is similar (point estimate $+0.006$, CI $[-0.041, 0.097]$ spanning zero with a worse sign; $p_{\text{Holm}} = 1.5 \times 10^{-4}$ again reflects only a weak directional regularity, not a meaningful change in magnitude), and is in fact worse for the CNN ($+0.080$) while better only for the Transformer (-0.119). Only the CCC loss — which penalizes scale and offset *inside* the objective rather than after it — delivers a consistent pooled calibrated-RMSE reduction (-0.057, $[-0.133, -0.027]$, CI excluding zero), and even that is driven by the CNN and Transformer. The confirmatory model agrees that the Pearson-loss effect on calibrated RMSE is null (-0.010, $t = -0.15$, CI spanning zero). In short, once predictions are correctly calibrated, the choice of correlation loss buys ranking quality, not magnitude accuracy; if calibrated magnitude is the explicit goal, the CCC loss is the better starting point.

5.4 Selection quality requires top-k and ranking metrics, and it is noisier than r

Because breeders act on the tails of the predicted distribution rather than on the average fit, we evaluated selection directly. Here the correlation-consistent losses again help on average — relative efficiency at the top 10% improves by $\Delta = +0.038$ to $+0.040$ across the three losses, top-10% overlap by $+0.014$ to $+0.018$, and NDCG@10 by $+0.016$ to $+0.017$ — and the gains track the predictive-ability gains, again concentrated in the CNN and Transformer and again negligible for the MLP. For the MLP specifically the selection effects of the Pearson loss are not distinguishable from zero (ΔNDCG@10 $= +0.0026$, $p_{\text{Holm}} = 0.28$; Δoverlap@10 $= +0.0015$, $p_{\text{Holm}} = 1.0$; both Holm-corrected, the convention we use throughout). Two cautions follow. First, a better r does not *guarantee* better selection of the extremes: the relationship between Δr and ΔNDCG@10 across cells is positive but loose, exactly as Blondel et al. (2015) and Ornella et al. (2014) warned, which is precisely why we report selection-aware metrics rather than inferring selection quality from r alone. Second, the most decision-relevant quantity is also the noisiest: selection differential at $k = 10\%$ has a positive pooled mean for the Pearson loss ($+0.25$) but with an interval that, while excluding zero ($[0.019, 0.612]$), is extremely wide because it depends on a handful of top individuals per fold. NDCG@k and relative efficiency are the better-behaved summaries and are what we recommend reporting. The broader point reinforces Waldmann (2019): the metric used to choose and evaluate a model changes the answer, and a genomic-selection study that reports only global r is not reporting the quantity it actually cares about.

5.5 Robustness under difficult splits is, at best, suggestive

We probed whether the benefit survives harder generalization regimes (leave-family-out on SoyNAM, cross-environment transfer on CIMMYT wheat, within-environment, and random K-fold) in a separate experiment. The honest summary is that the gain weakens, and in places reverses, under hard splits, and that the evidence base for these directional statements is thin and should be read as suggestive rather than confirmatory. The leave-family-out, random and within-environment contrasts each rest on only $n = 4$ datasets with no p-values, and the cross-environment contrast on $n = 12$ *ordered* environment pairs derived from the same four environments on the same 599 wheat lines, which are not mutually independent; no multiple-testing correction is applied across the roughly twenty split contrasts examined. With those caveats stated plainly: under leave-family-out CV the Pearson-loss predictive gain is essentially absent ($\Delta r = -0.0030$, $[-0.0106, 0.0010]$) and selection quality is, if anything, slightly degraded (ΔNDCG@10 $= -0.0042$, BCa CI $[-0.0089, -0.0005]$ excluding zero) — but this CI is bootstrapped over four paired cells with no p-value, so we read it as a suggestive signal of weakened benefit, not as established significant degradation. Under cross-environment transfer the predictive change is small and its inferential status is genuinely ambiguous: the point estimate is $\Delta r = +0.0098$ with a BCa CI of $[0.0001, 0.0204]$ that narrowly excludes zero, yet the paired Wilcoxon test is non-significant ($p = 0.151$). At $n = 12$ the discrete signed-rank test and the bootstrap CI simply disagree, and we do not resolve the disagreement in either direction; the cross-environment selection metric shows no benefit (ΔNDCG@10 $= -0.0012$, $p = 0.97$). The practically defensible reading is that the loss benefit is clearest under random partitioning and is not demonstrated to carry over to family- or environment-level extrapolation on the limited hard splits available here.

5.6 Recommendations for breeders and practitioners

We distil the above into concrete, defensible guidance.

1. **Match the loss to the architecture, not to fashion.** For linear predictors (GBLUP, ridge), keep their native estimators — they were the strongest methods in this study and have no loss to tune. Reserve the correlation-consistent loss for non-linear networks (CNN, Transformer), where it gave the clearest and most reliable gains; for a plain MLP it is essentially neutral and not worth the change.

2. **Always calibrate before reading predictions on their natural scale.** If the deliverable is a ranking, raw correlation-trained outputs suffice; if it is a breeding value, an RMSE, or anything compared against measured phenotypes, apply the closed-form affine map fit on held-out data. It preserves the ranking exactly whenever the fitted slope is positive (over 99% of folds) and is free.

3. **Report selection-aware metrics alongside r.** At minimum report NDCG@k and relative efficiency (or top-k overlap) at the selection intensities the program actually uses (here $k \in \{5, 10, 15, 20\}\%$). Treat selection differential as informative but high-variance.

4. **Prefer the CCC loss when calibrated scale and ranking both matter.** Among the objectives tested, the CCC loss was the *only* one to reduce pooled calibrated RMSE (CI excluding zero) while matching the other correlation-aware losses on ranking, giving a single training run that serves both magnitude and selection use-cases. The hybrid loss did *not* reliably improve pooled calibrated RMSE (point estimate +0.006, CI crossing zero); it lowered RMSE only for the Transformer and worsened it for the CNN, so its calibrated-RMSE benefit should not be generalized.

5. **Do not expect a loss change to substitute for a better model.** On additive, marker-dense traits the linear baselines remained ahead of all neural variants; the loss is a refinement, not a replacement for sound model choice.

5.7 Limitations and future work

Several limitations temper these conclusions. First, and most importantly for honest interpretation, our hyper-parameter search selected each network's architecture on full-data validation splits under a neutral MSE objective, after which the architecture was frozen and shared identically across all losses. Some test-fold genotypes' phenotypes can therefore have informed the architecture choice, so any *absolute* accuracy we report — raw and calibrated r values, the calibration-table RMSE figures, the batch-size correlations, and the neural rows of the mean-rank table — may be mildly, and we stress symmetrically, optimistically biased; none of these absolute numbers should be read as an unbiased estimate of out-of-sample accuracy. Because every loss and every calibration is compared *within* the same frozen architecture, and our inferential targets are always expressed as Δ versus MSE, this cannot bias the relative loss/calibration comparisons that constitute our claims; GBLUP, having no tuned hyper-parameters, is unaffected entirely, and the baseline-versus-network ranking is if anything conservative because the bias favours the networks. A fully nested HPO would remove even the cosmetic inflation of absolute numbers and is the natural next step.

Second, the gains, while statistically supported on random splits, are modest in absolute terms ($\Delta r \approx 0.04$ pooled) and uneven across architectures and species; they should not be oversold, and the pooled mean is computed over an architecturally unbalanced set of cells (CNN and MLP on all 101 trait–datasets, Transformer on only 49 across four species), with the largest per-architecture effect attaching to the smaller Transformer subset. Third, the most breeding-relevant metric, the selection differential, is estimated with high variance from few top individuals per fold, and larger panels or pooled multi-fold estimators would sharpen it. Fourth, our evidence on harder generalization regimes is limited. We caution against over-reading the LMM: its *scheme* factor has only two levels, "predefined" (the ten EasyGeSe species, their native five-fold CV partitions) and "random" (K-fold on SoyNAM and CIMMYT wheat), so it is fully confounded with dataset source and the $\beta_{\text{random}} = +0.072$ ($[-0.151, 0.296]$) coefficient contrasts those two dataset groups, *not* structured-versus-random splitting; moreover it is a main effect on the absolute metric, not a loss$\times$scheme interaction, so a null β_{random} says nothing about whether the loss benefit persists across schemes. The leave-family-out and cross-environment contrasts are not levels of this factor and are estimated only in the separate splits experiment, which — as discussed above — suggests the benefit weakens or reverses under family- and environment-level extrapolation rather than persisting. The number of genuinely independent hard splits is small ($n = 4$ families, $n = 12$ non-independent environment pairs), and establishing whether correlation-consistent training helps under realistic structured prediction will require substantially more, and more independent, hard-split data. Finally, our networks did not beat the linear baselines, so the practical ceiling on the value of any loss here is bounded by the predictors that benefit from it.

Future work follows directly. The group-aware variant of the loss (Section 2.4), which targets within-environment or within-family correlation rather than the pooled value, is the most promising extension for multi-environment breeding and was not exercised here. The mini-batch bias of the correlation loss (Proposition 3) motivates principled large-batch or statistic-accumulation training schemes. And a multi-trait formulation $\mathcal{L} = \sum_t w_t(1 - r_t)$ would let a single network serve correlated breeding objectives jointly. In each case the same discipline applies: align the training objective with the evaluation objective, calibrate when scale matters, and judge selection by selection metrics.

6 Conclusion

A standardized-MSE Pearson loss aligns neural-network training with the predictive-ability metric used throughout genomic prediction; because correlation is scale-free, affine calibration or a hybrid loss is required when calibrated phenotypic values matter; and for breeding decisions, global correlation should be complemented by top-k and ranking-aware metrics. Empirically, across 101 trait–datasets the benefit of correlation-consistent training is real but architecture-dependent — substantial for CNNs and Transformers, negligible for a well-tuned MLP — and on these benchmarks no neural loss unseated GBLUP or ridge on mean rank. The framework is thus a principled, well-characterized tool to be applied judiciously, not a universal replacement for MSE.

Data and code availability

All datasets are public: EasyGeSe (Zenodo record 15348871), the CIMMYT wheat panel (`BGLR` R package) and SoyNAM (`SoyNAM` R package). Code, fixed cross-validation partitions, tuned configurations and a pinned environment are released as the `ccgp` package, with source at `https://github.com/GBeurier/selgen-loss` (a versioned archive will be deposited at Zenodo, DOI to be assigned). The sorghum case-study data (Section 4.8) are proprietary and not redistributed; all other datasets are public and download automatically.

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